CAC2026 - Data Challenge

Olive Oil Sensory Defects Detection

We are pleased to announce a **chemometrics data challenge** in the context of the XX Chemometrics in Analytical Chemistry conference.

Context

The quality evaluation and categorisation of olive oils is performed by official sensory panels, which determine that for instance, an olive oil can be categorised as extra virgin because it has no sensory defects, among other evaluated parameters (fruitiness, chemical composition…). Therefore, it is really relevant for the industry to develop analytical methods to detect these defects early, methods that go through instrumental analysis and chemometrics.

In this regard, participants will work with an olive oil dataset based on instrumental fingerprints acquired by **headspace-mass spectrometry (HS-MS), mid-infrared spectroscopy (MIR)** and **UV–visible spectrophotometry (UV–vis)**. The reference values (**detection of the musty defect**) are derived from sensory assessment by an official taste panel. The dataset is associated with the following reference paper:

Borràs et al., Food Chemistry, 2016.

DOI: **10.1016/j.foodchem.2016.02.038**

Challenge objective

Participants are asked to develop **one predictive model** to classify olive oil samples according to the presence or absence of the musty defect.

This target has been selected because it is chemically meaningful, relevant to olive oil quality assessment, and expected to allow good modelling performance without being trivial.

Participants may use ANY modelling strategy.

Available data

Participants will receive datasets including:

- HS-MS data
- ATR-FTIR spectral data
- UV–vis spectral data
- Training class labels
- Training class metadata
- Test samples without labels nor metadata

The use of one, two or all three data blocks is allowed. Participants have full freedom to decide how to preprocess, combine, select or model the data.

Evaluation

Models will be evaluated solely based on the performance on the provided test set.

The evaluation criterion will be the F1 metric:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

where:

$$Precision = \frac{TP}{TP + FP}$$
$$Recall = \frac{TP}{TP + FN}$$

The final ranking will consider predictive performance. The organisers may also highlight particularly interesting approaches, even if they are not the top-ranked model. In the event of a tie, the principle of parsimony will apply: the simpler model (determined by the scientific committee of the conference) will be ranked higher.

Important clarifications

- Samples in the calibration set and the test set are totally independent.
- The **replicates are analytical/instrumental** replicates, therefore, replicate 2 for UV-Vis is not the same subsample as replicate 2 for ATR-FTIR, for instance.
- There **may be outliers in the calibration** set, in the sample-level or in the measurement-level.
- There are no sample-level outliers in the test set, even though the replicate measurements may differ. Therefore, **all test samples must be classified**.
- **Predictions** must be made **at the sample-level**, even though replicate measurements are provided.
- External information that reveals or infers the identity of test samples is not allowed.
- The dataset must not be redistributed or uploaded to public repositories.
- Submitted results may be used by the organisers for conference presentations, reports, rankings, proceedings or future scientific dissemination, using anonymised data, non commercially.

Submission requirements

Each participant or team must submit:

1. **Modelling pipeline description**
2. **Predicted class for each test sample**
3. **Declaration of data use and rights**, confirming acceptance of the challenge conditions.

The submission will be made through an online form before the deadline:

CAC2026 Data challenge – Fill in form

Submission deadline: 26th June, Friday, 2026

Late submissions will not be included in the official ranking.

Prize

A prize (to be determined) will be awarded to the best-performing participant or team.

Further details will be announced before the challenge deadline.

Who can participate?

The challenge is aimed at conference participants, but open to anyone with an interest in chemometrics, analytical chemistry, spectroscopy, sensory analysis, food authentication, machine learning or data fusion.

Participation may be individual or in teams.

Why participate?

This challenge offers an opportunity to test chemometric strategies on a real analytical dataset involving complementary instrumental techniques and sensory reference data. It is also a chance to compare modelling approaches, explore data fusion, and discuss the practical difficulties of translating instrumental fingerprints into sensory quality predictions.

Ultimate aim of the activity

The scientific aim of this challenge is to use a shared olive oil dataset as a common benchmark to compare how different chemometricians approach the same modelling problem, from preprocessing and variable selection to data fusion, validation and final classification strategy. Beyond identifying the best-performing model, the challenge seeks to provide an overview of current chemometric practice: which methods are preferred, how modelling decisions are justified, how uncertainty and class imbalance are handled, and how instrumental fingerprints are translated into chemically meaningful predictions. By collecting and comparing diverse solutions on the same data, the challenge will offer a practical snapshot of the state of the art in chemometric modelling and encourage discussion on good practices, reproducibility and methodological transparency.